

ACFI315 Principles of finance with Excel

Week 2 Forecasting in Excel (Linear Relationship)

A number of data files have been uploaded to VITAL.

1. LVMH Product Data

Research questions:

- (a) Find total revenue for 2000-2018 for each product segment.
- (b) Find the share of total revenue for each product segment.
- (c) Find revenue growth for each year.
- (d) Use Sparklines to illustrate the change in revenue
- (e) Use a doughnut diagram to illustrate the change in share of revenue from 2000 to 2018 (inner ring 2000, outer ring 2018).
- (f) Copy the “Wines and Spirits data” to Sheet 2 so that it is ordered vertically not horizontally. Hint: Copy & Transpose Values: CTRL + C (to copy to clipboard); CTRL + ALT + V (to bring up Paste Special Menu); E (for the Transpose sub-command); Enter (to paste)
- (g) Draw a line chart of Wines and Spirits Data.
- (h) Fit a trend line and show the equation.
- (i) Use **INTERCEPT**, **SLOPE** and **RSQ** to find the intercept, slope and r-squared values.
- (j) Use **FORECAST** to forecast the “Wines and Spirits” sales in years 20, 21 and 22 (2018, 2019 and 2020).
- (k) Repeat the step above using **TREND**.

2. Apple Sales Data

Research questions:

- (a) Find (i) Gross Margin; (ii) Total Operating Expenses; (iii) Operating Income; (iv) Income before tax and (v) Net Income.
- (b) Find (i) effective tax rate; (ii) R&D as percent of sales, (iii) Selling, General and Admin as percent of sales and (iv) other Income as % of operating income.
- (c) Find (i) Growth rate in sales; (ii) gross profit margin and (iii) net profit margin.
- (d) Use the **DATE**, **YEAR**, **MONTH** and **DAY** functions to fill cells N1 with the date of 30/09/2019.
- (e) Assuming that sales continue to grow for the next five years at the same rate that they grew from 2017 to 2018 and that the effective tax rate, R&D as percent of sales, Selling, General and Admin as percent of sales, other Income as % of OI and gross profit margin remain constant, forecast the income statement for the years 2019.